

(bars: lengths by R , gain by a_{nom} ; hats on estimates; $e_\theta = \theta - \hat{\theta}$; $\bar{w}_d[k]$ commanded height, $\Delta\bar{w}_d[k] = \bar{w}_d[k+1] - \bar{w}_d[k]$; $\bar{a}' := \bar{a}'_{\text{true}}(\bar{w}_d[k])$, $\bar{a}'' := \bar{a}''_{\text{true}}(\bar{w}_d[k])$, both exogenous; $\Delta\bar{w}[k] := \bar{w}[k+1] - \bar{w}[k]$; $\Delta\hat{w}[k] := \Delta\bar{w}_d[k] + (1 - \lambda_c) \delta\hat{w}_3[k]$; $e_{\Delta\bar{w}} := \Delta\bar{w} - \Delta\hat{w}$; $\mathbb{E}[\cdot]$ mean over noise realizations; $P_{ij} = \mathbb{E}[e_i e_j]$; $d = 2$)

1 True dynamics

$$\begin{aligned}
\bar{w}[k] &= \bar{w}_d[k] - \delta\bar{w}_3[k] \\
\delta\bar{w}_3[k+1] &= \lambda_c \delta\bar{w}_3[k] - F_{dw}[k] e_{\bar{a}_w}[k] - \varepsilon_w[k] \\
\Delta\bar{w}[k] &= \Delta\bar{w}_d[k] + (1 - \lambda_c) \delta\bar{w}_3[k] + F_{dw}[k] e_{\bar{a}_w}[k] + \varepsilon_w[k] \\
\bar{a}_w[k+1] - \bar{a}_w[k] &= \bar{a}'_{\text{true}}(\bar{w}[k]) \Delta\bar{w}[k] + \frac{1}{2} \bar{a}''_{\text{true}}(\bar{w}[k]) \Delta\bar{w}[k]^2 + \mathcal{O}(\Delta\bar{w}^3) \\
\bar{a}'_{\text{true}}(\bar{w}[k]) &= \bar{a}'_{\text{true}}(\bar{w}_d[k] - \delta\bar{w}_3[k]) = \bar{a}' - \bar{a}'' \delta\bar{w}_3[k] + \mathcal{O}(\delta\bar{w}_3^2) \\
\bar{a}''_{\text{true}}(\bar{w}[k]) &= \bar{a}'' + \mathcal{O}(\delta\bar{w}_3)
\end{aligned}$$

(first line: $\bar{a}_{\text{true}}$ expanded about $\bar{w}[k]$ in $\Delta\bar{w}$; second line: $\bar{a}'_{\text{true}}$ expanded about $\bar{w}_d[k]$ in $-\delta\bar{w}_3$)

$$\bar{a}_w[k+1] - \bar{a}_w[k] = (\bar{a}' - \bar{a}'' \delta\bar{w}_3[k]) \Delta\bar{w}[k] + \frac{1}{2} \bar{a}'' \Delta\bar{w}[k]^2$$

$$\begin{aligned}
F_{dw}[k] &= \bar{f}_{dw}[k] + (1 - \lambda_c) \sum_{i=1}^d \bar{f}_{dw}[k-i] \\
\varepsilon_w[k] &= \bar{a}_w[k] \bar{f}_T[k] + (1 - \lambda_c) \sum_{i=1}^d \bar{a}_w[k-i] \bar{f}_T[k-i] - (1 - \lambda_c) n_w[k]
\end{aligned}$$

2 Estimator

$$\begin{aligned}
\delta\hat{w}_3[k+1] &= \lambda_c \delta\hat{w}_3[k] \\
\Delta\hat{w}[k] &= \Delta\bar{w}_d[k] + (1 - \lambda_c) \delta\hat{w}_3[k] \\
\hat{a}_w[k+1] - \hat{a}_w[k] &= \bar{a}' \Delta\hat{w}[k] + \frac{1}{2} \bar{a}'' \Delta\hat{w}[k]^2
\end{aligned}$$

3 Error increment

$$\begin{aligned} e_{\Delta\bar{w}}[k] &= \Delta\bar{w}[k] - \Delta\hat{w}[k] = (1 - \lambda_c) e_{\delta\bar{w}_3}[k] + F_{dw}[k] e_{\bar{a}_w}[k] + \varepsilon_w[k] \\ \delta\bar{w}_3[k] &= \delta\hat{w}_3[k] + e_{\delta\bar{w}_3}[k] \end{aligned}$$

$$\begin{aligned} e_{\bar{a}_w}[k+1] - e_{\bar{a}_w}[k] &= (\bar{a}_w[k+1] - \bar{a}_w[k]) - (\hat{a}_w[k+1] - \hat{a}_w[k]) \\ &= (\bar{a}' - \bar{a}'' \delta\bar{w}_3)(\Delta\hat{w} + e_{\Delta\bar{w}}) + \frac{1}{2} \bar{a}'' (\Delta\hat{w} + e_{\Delta\bar{w}})^2 - \bar{a}' \Delta\hat{w} - \frac{1}{2} \bar{a}'' \Delta\hat{w}^2 \\ &= \bar{a}' e_{\Delta\bar{w}} - \bar{a}'' \delta\bar{w}_3 \Delta\hat{w} - \bar{a}'' \delta\bar{w}_3 e_{\Delta\bar{w}} + \bar{a}'' \Delta\hat{w} e_{\Delta\bar{w}} + \frac{1}{2} \bar{a}'' e_{\Delta\bar{w}}^2 + \mathcal{O}(3) \end{aligned}$$

4 Mean

$$\begin{aligned} \mathbb{E}[e_{\delta\bar{w}_3}] &= \mathbb{E}[e_{\bar{a}_w}] = \mathbb{E}[\varepsilon_w] = 0 \\ \mathbb{E}[e_{\delta\bar{w}_3}^2] &= P_{33}, \quad \mathbb{E}[e_{\delta\bar{w}_3} e_{\bar{a}_w}] = P_{34}, \quad \mathbb{E}[e_{\bar{a}_w}^2] = P_{44}, \quad \mathbb{E}[\varepsilon_w^2] = Q_{33} \\ \mathbb{E}[\varepsilon_w e_{\delta\bar{w}_3}] &= \mathbb{E}[\varepsilon_w e_{\bar{a}_w}] = 0 \end{aligned}$$

$$\begin{aligned} \mathbb{E}[\bar{a}' e_{\Delta\bar{w}}] &= 0 \\ \mathbb{E}[-\bar{a}'' \delta\bar{w}_3 \Delta\hat{w}] &= -\bar{a}'' \delta\hat{w}_3 \Delta\hat{w} \\ \mathbb{E}[-\bar{a}'' \delta\bar{w}_3 e_{\Delta\bar{w}}] &= -\bar{a}'' [(1 - \lambda_c) P_{33} + F_{dw} P_{34}] \\ \mathbb{E}[\bar{a}'' \Delta\hat{w} e_{\Delta\bar{w}}] &= 0 \\ \mathbb{E}[\frac{1}{2} \bar{a}'' e_{\Delta\bar{w}}^2] &= \frac{1}{2} \bar{a}'' [(1 - \lambda_c)^2 P_{33} + 2(1 - \lambda_c) F_{dw} P_{34} + F_{dw}^2 P_{44} + Q_{33}] \end{aligned}$$

$$\mathbb{E}[e_{\bar{a}_w}[k+1]] - \mathbb{E}[e_{\bar{a}_w}[k]] = \bar{a}'' \left\{ -\delta\hat{w}_3 \Delta\hat{w} - \frac{1}{2} (1 - \lambda_c^2) P_{33} - \lambda_c F_{dw} P_{34} + \frac{1}{2} F_{dw}^2 P_{44} + \frac{1}{2} Q_{33} \right\}$$

5 Estimator with the mean term

$$\begin{aligned} \hat{a}_w[k+1] &= \hat{a}_w[k] + \bar{a}' \Delta\hat{w}[k] + \frac{1}{2} \bar{a}'' \Delta\hat{w}[k]^2 \\ &\quad + \bar{a}'' \left\{ -\delta\hat{w}_3[k] \Delta\hat{w}[k] - \frac{1}{2} (1 - \lambda_c^2) P_{33}[k] - \lambda_c F_{dw}[k] P_{34}[k] \right. \\ &\quad \left. + \frac{1}{2} F_{dw}[k]^2 P_{44}[k] + \frac{1}{2} Q_{33}[k] \right\} \end{aligned}$$

$$\mathbb{E}[e_{\bar{a}_w}[k+1]] - \mathbb{E}[e_{\bar{a}_w}[k]] = 0$$

(the braced term is a deterministic input to the predict; it is not differentiated and does not enter F_e , H , Q)

6 Jacobians

$$\frac{\partial \hat{a}_w[k+1]}{\partial \delta \hat{w}_3[k]} = (1 - \lambda_c)(\bar{a}' + \bar{a}'' \Delta \hat{w}[k]) \rightarrow (1 - \lambda_c) \bar{a}'$$

$$\frac{\partial \hat{a}_w[k+1]}{\partial \hat{a}_w[k]} = 1 + F_{dw}[k](\bar{a}' + \bar{a}'' \Delta \hat{w}[k]) \rightarrow 1 + F_{dw}[k] \bar{a}'$$

(limits at $\Delta \hat{w} \rightarrow 0$; the F_{dw} term is the truth's $F_{dw} e_{\bar{a}_w}$ inside $\Delta \bar{w}$, written as $F_{dw}(\bar{a}_w - \hat{a}_w)$)

7 Error dynamics

$$e_{\delta \bar{w}_1}[k+1] = e_{\delta \bar{w}_2}[k]$$

$$e_{\delta \bar{w}_2}[k+1] = e_{\delta \bar{w}_3}[k]$$

$$e_{\delta \bar{w}_3}[k+1] = \lambda_c e_{\delta \bar{w}_3}[k] - F_{dw}[k] e_{\bar{a}_w}[k] - \varepsilon_w[k]$$

$$\begin{aligned} e_{\bar{a}_w}[k+1] &= (1 - \lambda_c)(\bar{a}' + \bar{a}'' \Delta \hat{w}) e_{\delta \bar{w}_3}[k] + \left[1 + F_{dw}(\bar{a}' + \bar{a}'' \Delta \hat{w})\right] e_{\bar{a}_w}[k] \\ &\quad + (\bar{a}' + \bar{a}'' \Delta \hat{w}) \varepsilon_w[k] \\ &\quad - \bar{a}'' \delta \hat{w}_3 \Delta \hat{w} - \bar{a}'' \Delta \hat{w} e_{\delta \bar{w}_3} - \bar{a}'' \delta \hat{w}_3 e_{\Delta \bar{w}} - \bar{a}'' e_{\delta \bar{w}_3} e_{\Delta \bar{w}} + \frac{1}{2} \bar{a}'' e_{\Delta \bar{w}}^2 \end{aligned}$$

$$\mathbf{e}[k+1] = F_e[k] \mathbf{e}[k] + \mathbf{q}[k], \quad F_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & \lambda_c & -F_{dw} \\ 0 & 0 & (1 - \lambda_c)(\bar{a}' + \bar{a}'' \Delta \hat{w}) & 1 + F_{dw}(\bar{a}' + \bar{a}'' \Delta \hat{w}) \end{bmatrix}$$

$$\mathbf{q}[k] = \begin{bmatrix} 0 \\ 0 \\ -\varepsilon_w \\ (\bar{a}' + \bar{a}'' \Delta \hat{w}) \varepsilon_w \end{bmatrix}$$

$$P^f[k+1] = F_e[k] P[k] F_e[k]^\top + Q[k]$$

8 Measurement and update

$$\nabla_d \bar{w}_d[k] = \bar{w}_d[k] - \bar{w}_d[k-d]$$

$$y_1[k] = \delta \bar{w}_1[k] + n_w[k]$$

$$y_2[k] = \bar{a}_w[k-d] + n_a[k], \quad \bar{a}_w[k-d] = \bar{a}_w[k] - \bar{a}' \nabla_d \bar{w}_d[k]$$

$$H = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$e_{y_1}[k] = e_{\delta \bar{w}_1}[k] + n_w[k], \quad e_{y_2}[k] = e_{\bar{a}_w}[k] + n_a[k]$$

$$\begin{aligned}
L[k] &= P^f H^\top (H P^f H^\top + R)^{-1} \\
\hat{\mathbf{x}}^+[k] &= \hat{\mathbf{x}}^-[k] + L[k] \mathbf{e}_y[k] \\
P[k] &= (I - L[k] H) P^f[k]
\end{aligned}$$

9 Noise

$$\begin{aligned}
\kappa_T &= \frac{4k_B T a_o}{R}, \quad \sigma_{n_w}^2 = \frac{\sigma_n^2}{R^2} \\
Q_{33} &= \kappa_T \left(\hat{a}_w[k] + (1 - \lambda_c)^2 (\hat{a}_w[k-1] + \hat{a}_w[k-2]) \right) + (1 - \lambda_c)^2 \sigma_{n_w}^2 \\
Q_{34} = Q_{43} &= -(\bar{a}' + \bar{a}'' \Delta \hat{w}) Q_{33} \\
Q_{44} &= (\bar{a}' + \bar{a}'' \Delta \hat{w})^2 Q_{33} \\
R &= \text{diag}(R_1, R_2), \quad R_1 = \sigma_{n_w}^2, \quad R_2 = 2a_{\text{cov}}^2 IF_{\text{var}}(\hat{a}_w) (\hat{a}_w + \xi)^2
\end{aligned}$$

(IF_{var} , ξ as in 0902_formC_aptrue_4state.tex)

